

grapes are grafted onto rootstocks of varieties that produce inferior grapes but are more resistant to certain soil pathogens. The genes of the scion determine the quality of the fruit. During grafting, a callus first forms between the adjoining cut ends of the scion and stock; cell differentiation then completes the functional unification of the grafted individuals.

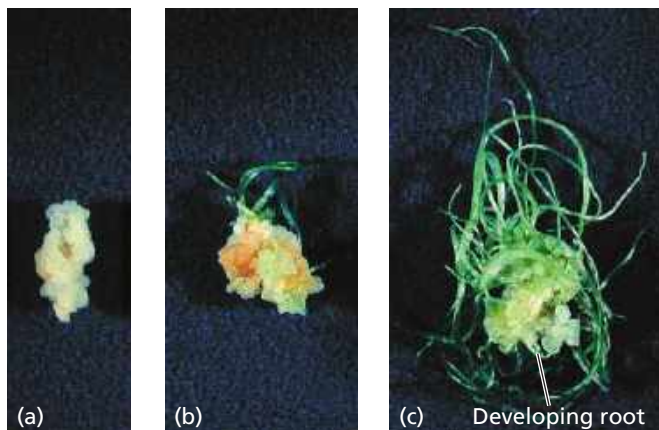
Test-Tube Cloning and Related Techniques

Plant biologists have adopted *in vitro* methods to clone plants for research or horticulture. Whole plants can be obtained by culturing small pieces of tissue from the parent plant on an artificial medium containing nutrients and hormones. The cells or tissues can come from any part of a plant, but growth may vary depending on the plant part, species, and artificial medium. In some media, the cultured cells divide and form a callus of undifferentiated totipotent cells (**Figure 38.15a**). When the concentrations of hormones and nutrients are manipulated appropriately, a callus can sprout shoots and roots with fully differentiated cells (**Figure 38.15b and c**). If desired, the cloned plantlets can then be transferred to soil, where they continue their growth.

Plant tissue culture is important in eliminating weakly pathogenic viruses from vegetatively propagated varieties. Although the presence of weak viruses may not be obvious, yield or quality may be substantially reduced as a result of infection. Strawberry plants, for example, are susceptible to more than 60 viruses, and typically the plants must be replaced each year because of viral infection. However, since the apical meristems are often virus-free, they can be excised and used to produce virus-free material for tissue culture.

Plant tissue culture also facilitates genetic engineering, the deliberate modification of the characteristics of an organism by manipulating its genetic material (see Concept 20.1). Many genetic engineering techniques require single plant cells as the starting material. Test-tube culture makes it possible to regenerate plants from a single genetically engineered

▼ **Figure 38.15 Cloning a garlic plant.** (a) A root from a garlic clove gave rise to this callus culture, a mass of undifferentiated totipotent cells. (b and c) The differentiation of a callus into a plantlet depends on the nutrient levels and hormone concentrations in the artificial medium, as can be seen in these cultures grown for different lengths of time.



plant cell. In the next section, we'll take a closer look at some of the promises and challenges surrounding the use of genetically modified plants in agriculture.

CONCEPT CHECK 38.2

1. What are three ways that flowering plants avoid self-fertilization?
2. The seedless banana, the world's most popular fruit, is losing the battle against two fungal epidemics. Why do such epidemics generally pose a greater risk to asexually propagated crops?
3. Self-fertilization, or selfing, seems to have obvious disadvantages as a reproductive "strategy" in nature, and it has even been called an "evolutionary dead end." So it is surprising that about 20% of angiosperm species primarily rely on selfing. Suggest a reason why selfing might be advantageous and yet still be an evolutionary dead end.

For suggested answers, see Appendix A.

CONCEPT 38.3

People modify crops by breeding and genetic engineering

People have intervened in the reproduction and genetic makeup of plants since the dawn of agriculture. Maize, for example, owes its existence to humans. Left on its own in nature, maize would soon become extinct for the simple reason that it cannot spread its seeds. Maize kernels are not only permanently attached to the central axis (the "cob") but also permanently protected by tough, overlapping leaf sheathes (the "husk") (**Figure 38.16**). These attributes arose by artificial selection by humans. Despite having no understanding of the scientific principles underlying plant breeding, early farmers domesticated most of our crop species over a relatively short period about 10,000 years ago.

▼ Figure 38.16 Maize: a product of artificial selection.

Modern maize (bottom) was derived from teosinte (top). Teosinte kernels are tiny, and each row has a husk that must be removed to get at the kernel. The seeds are loose at maturity, allowing dispersal, which probably made harvesting difficult for early farmers. Neolithic farmers selected seeds from plants with larger cob and kernel size as well as the permanent attachment of seeds to the cob and the encasing of the entire cob by a tough husk.



➔ Mastering Biology HHMI Video: Popped Secret: The Mysterious Origin of Corn

